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(54) **SAFETY CAP RETAINER**

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(57) **ABSTRACT**

Related U.S. Application Data

(60) Provisional application No. 63/612,213, filed on Dec. 19, 2023.

Methods, compositions and combinations for securing a protective rebar cover to the end of a segment of exposed reinforcing bar.

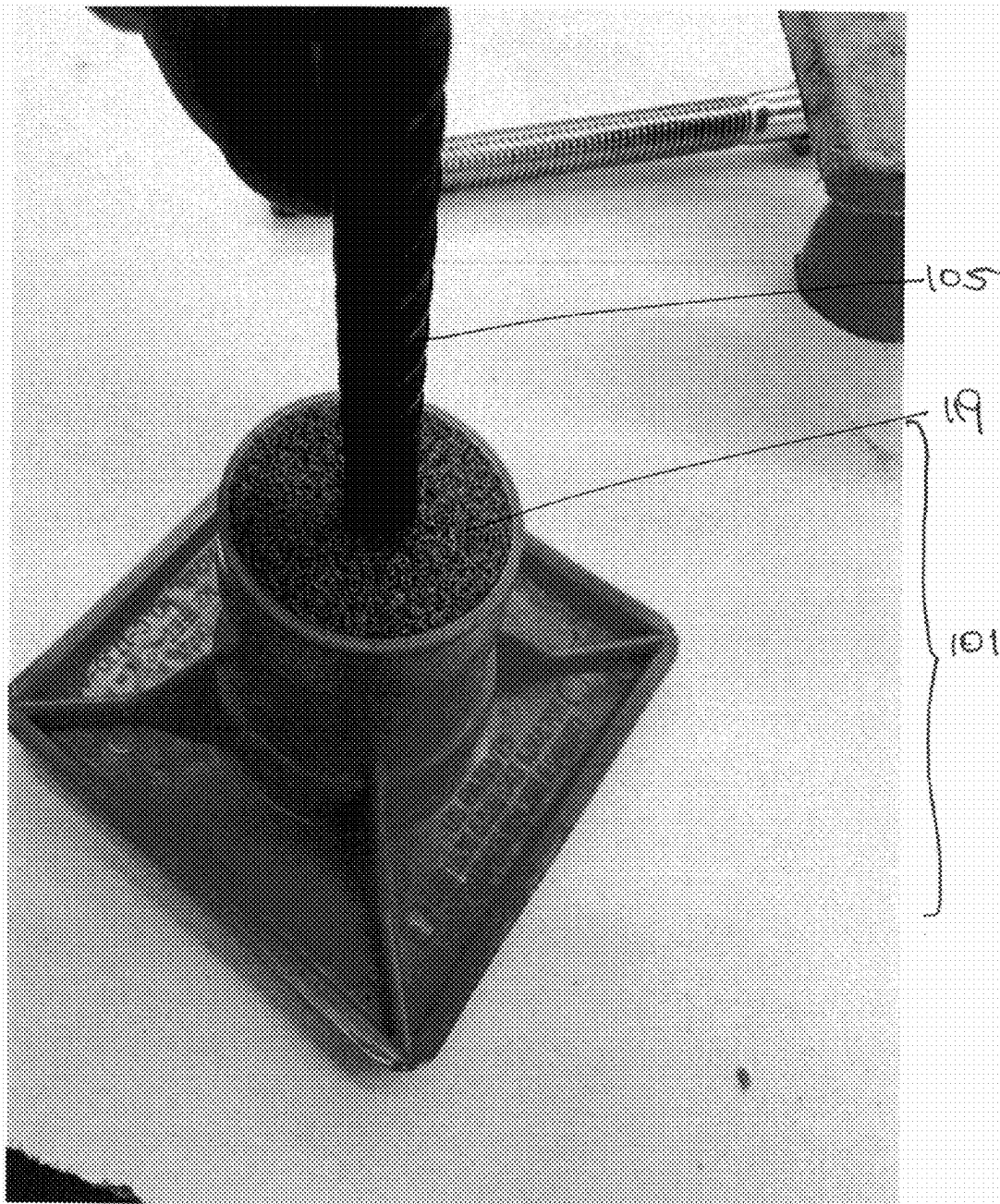


Fig 1

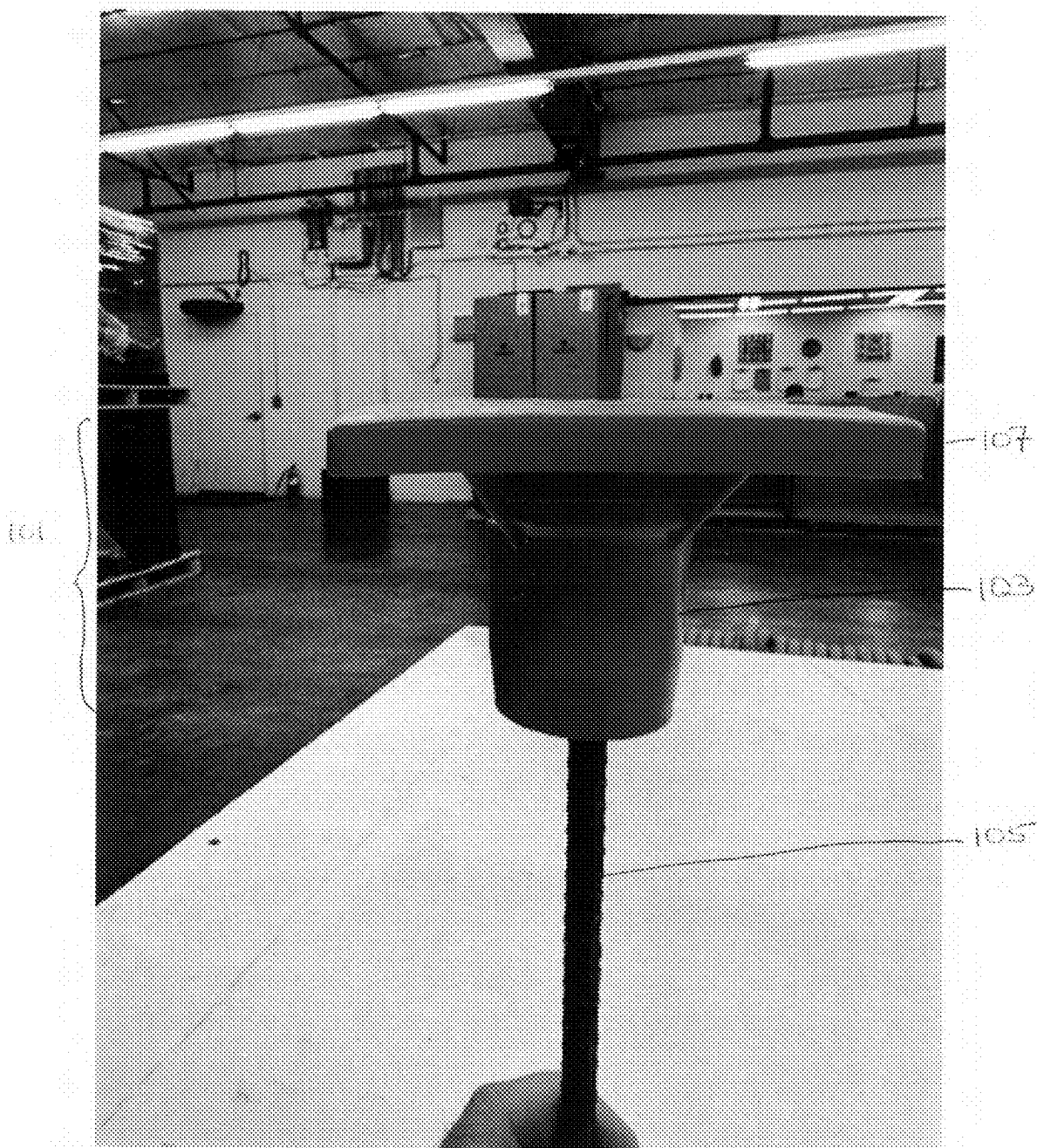


Fig. 2

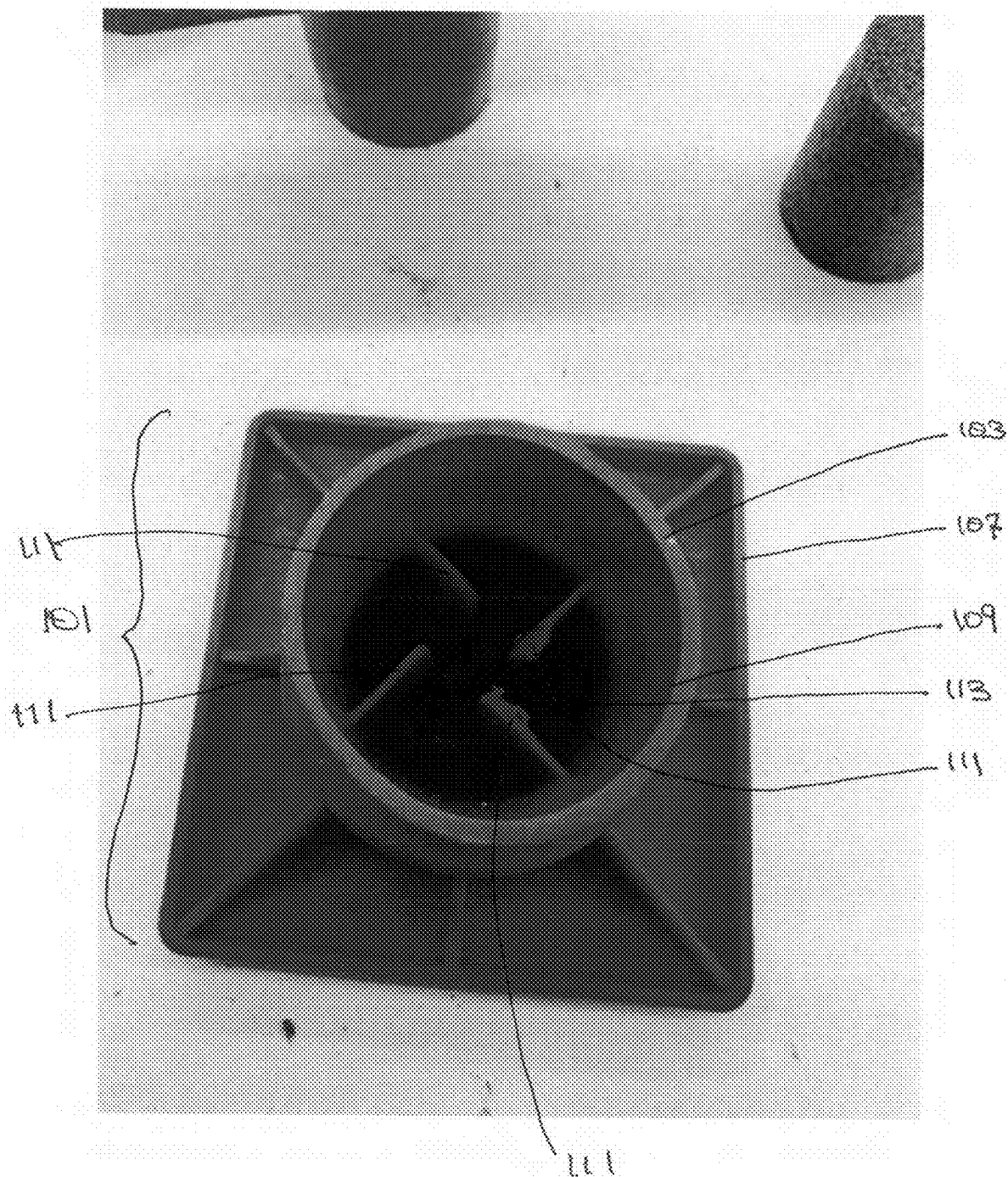
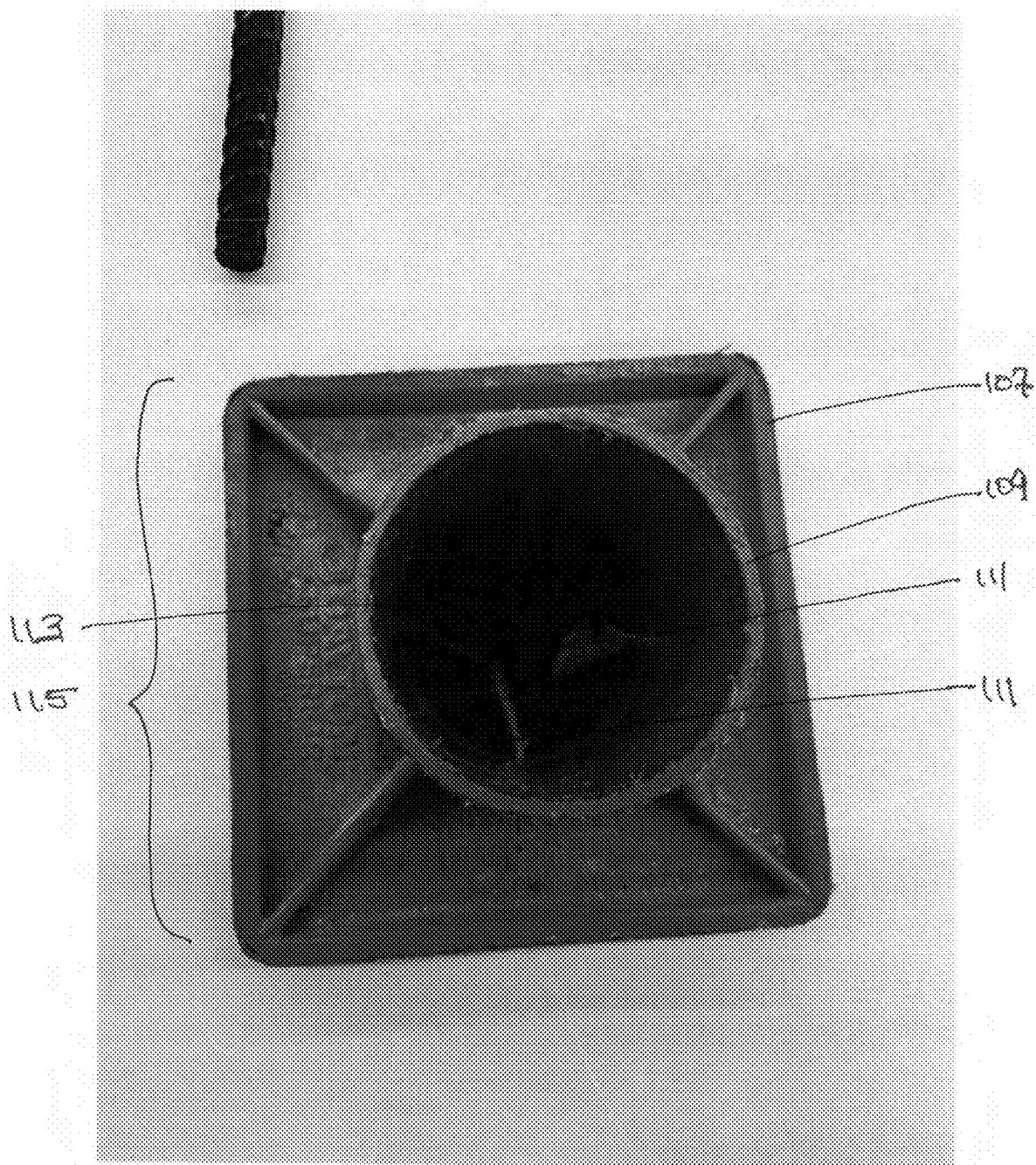


Fig 3



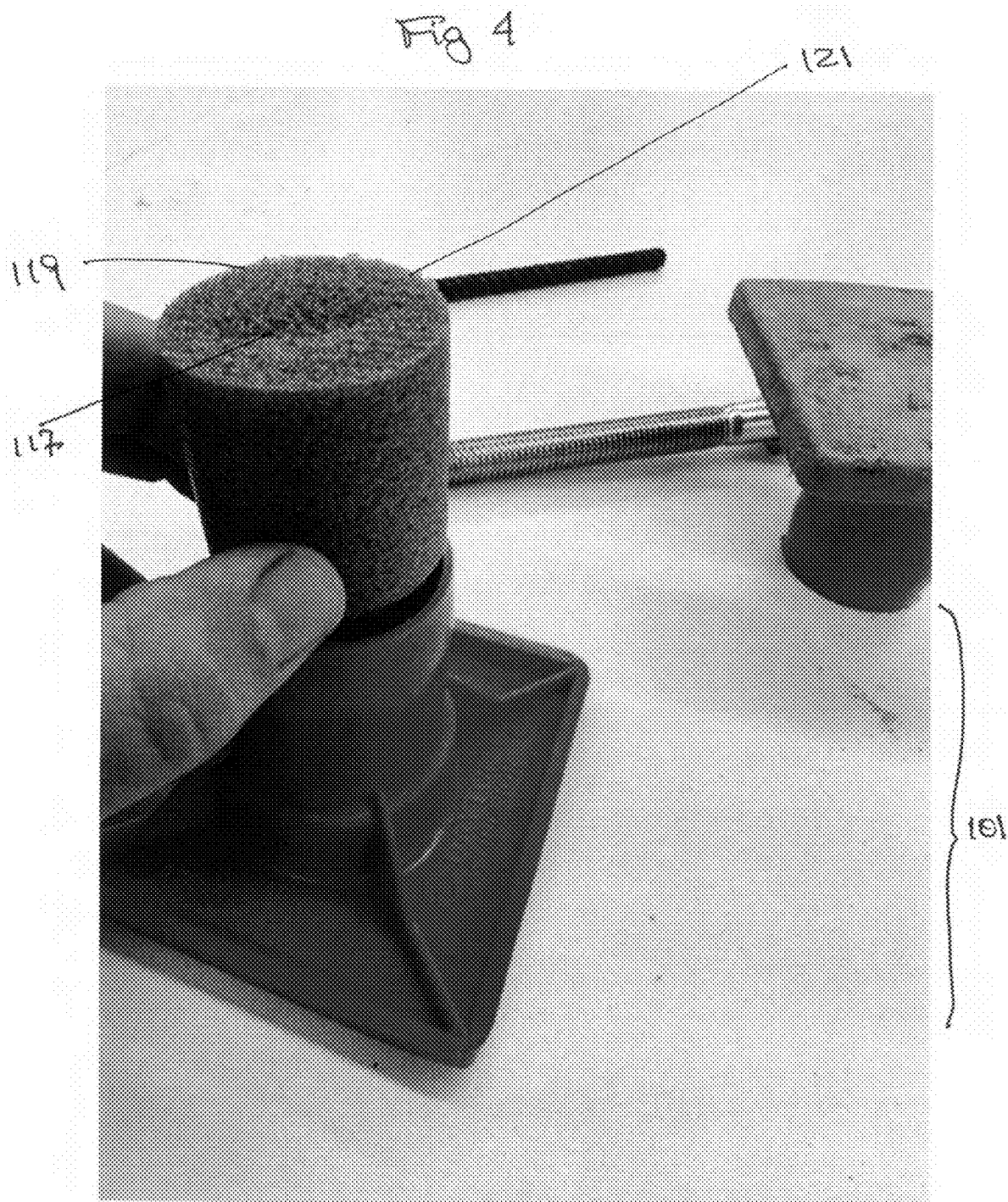


Fig 5

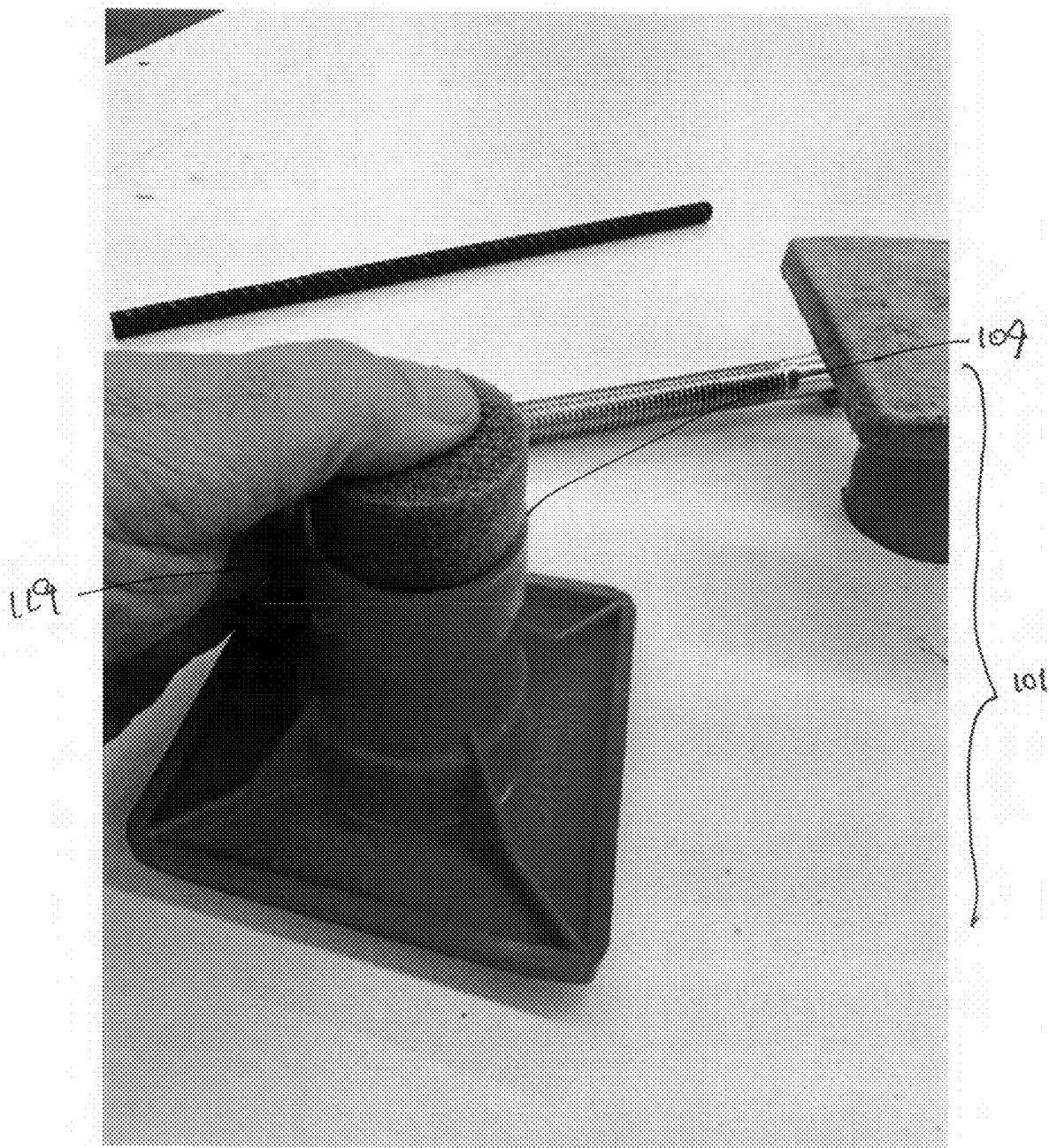


Fig. 6

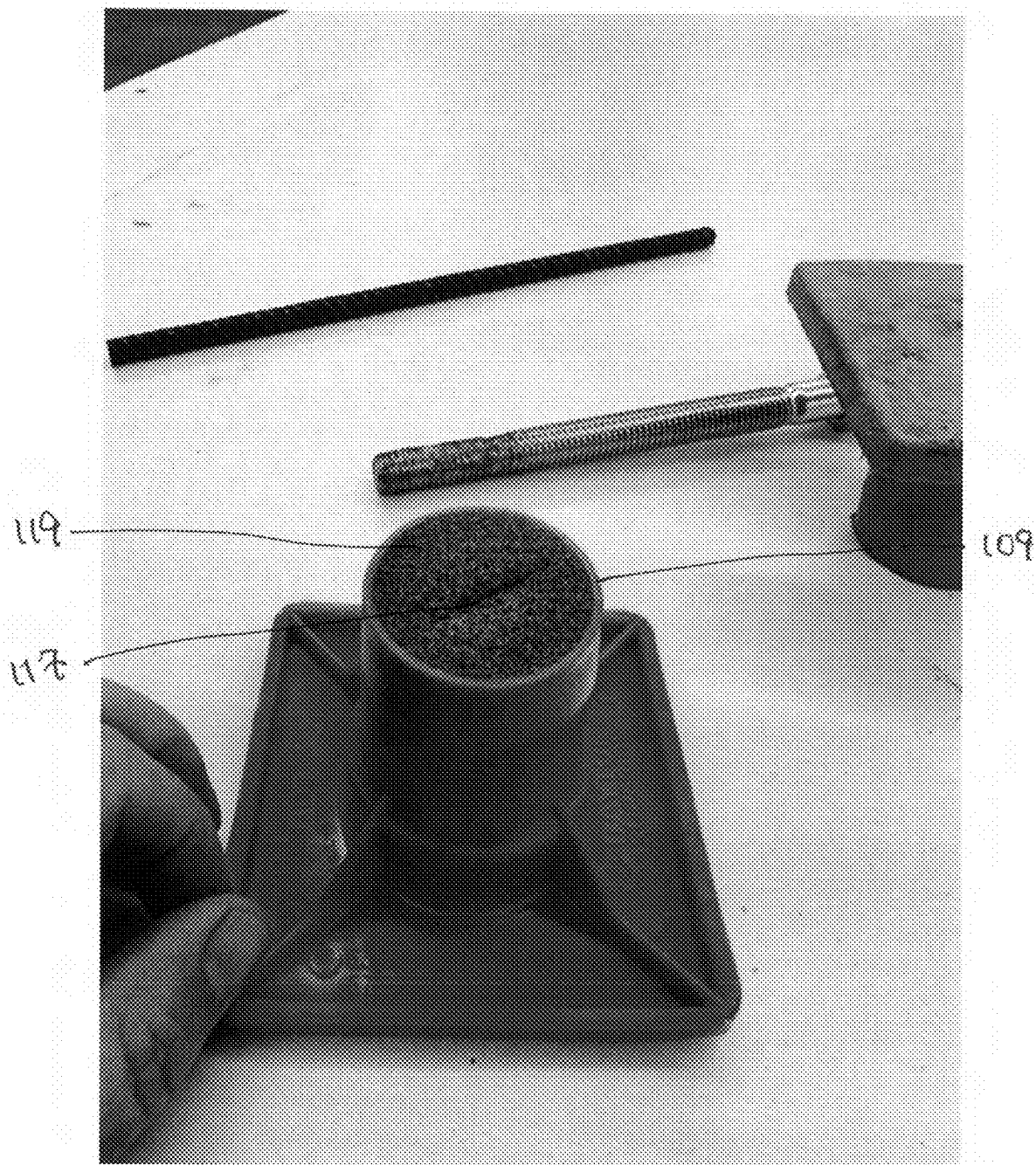


Fig. 7

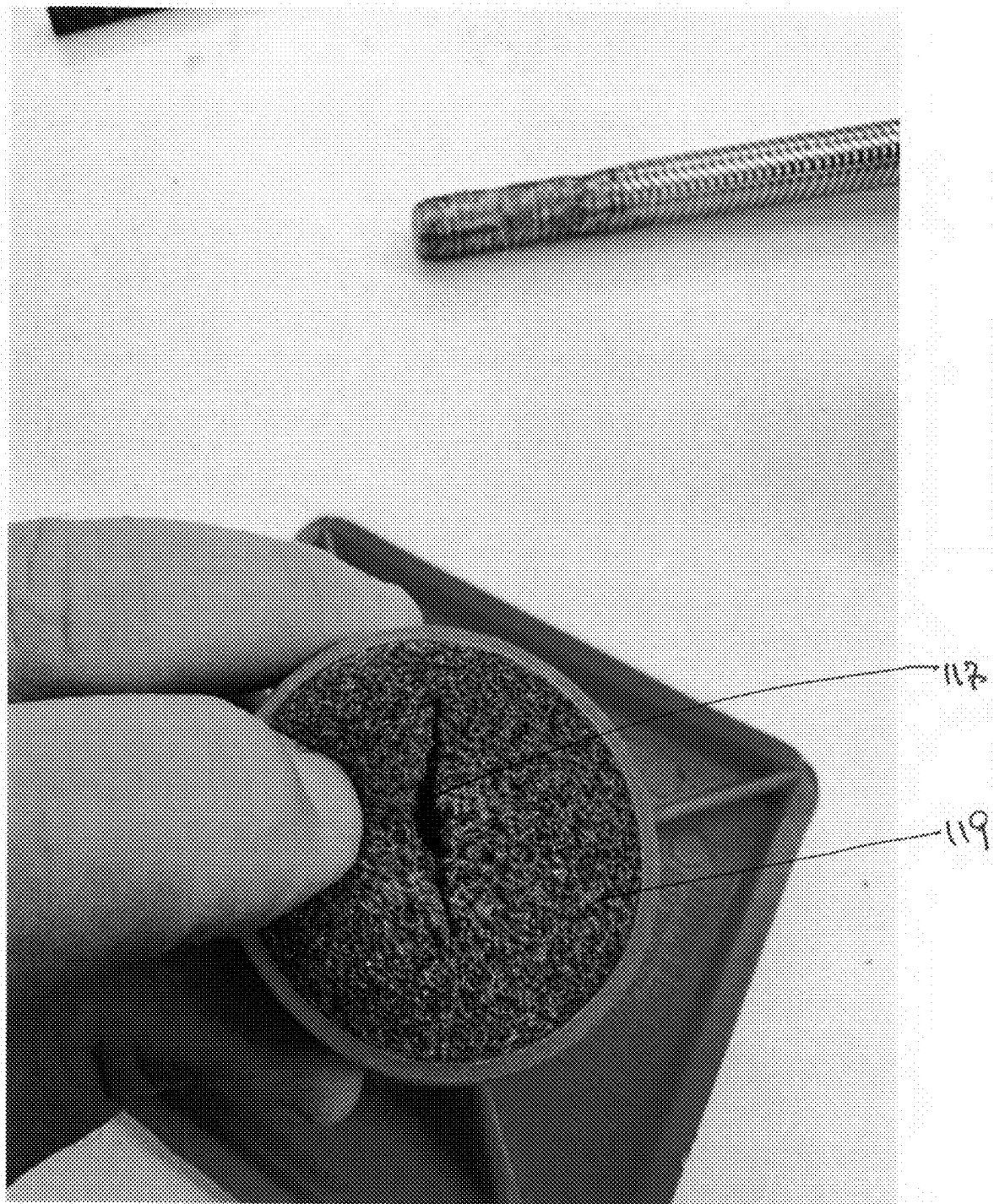
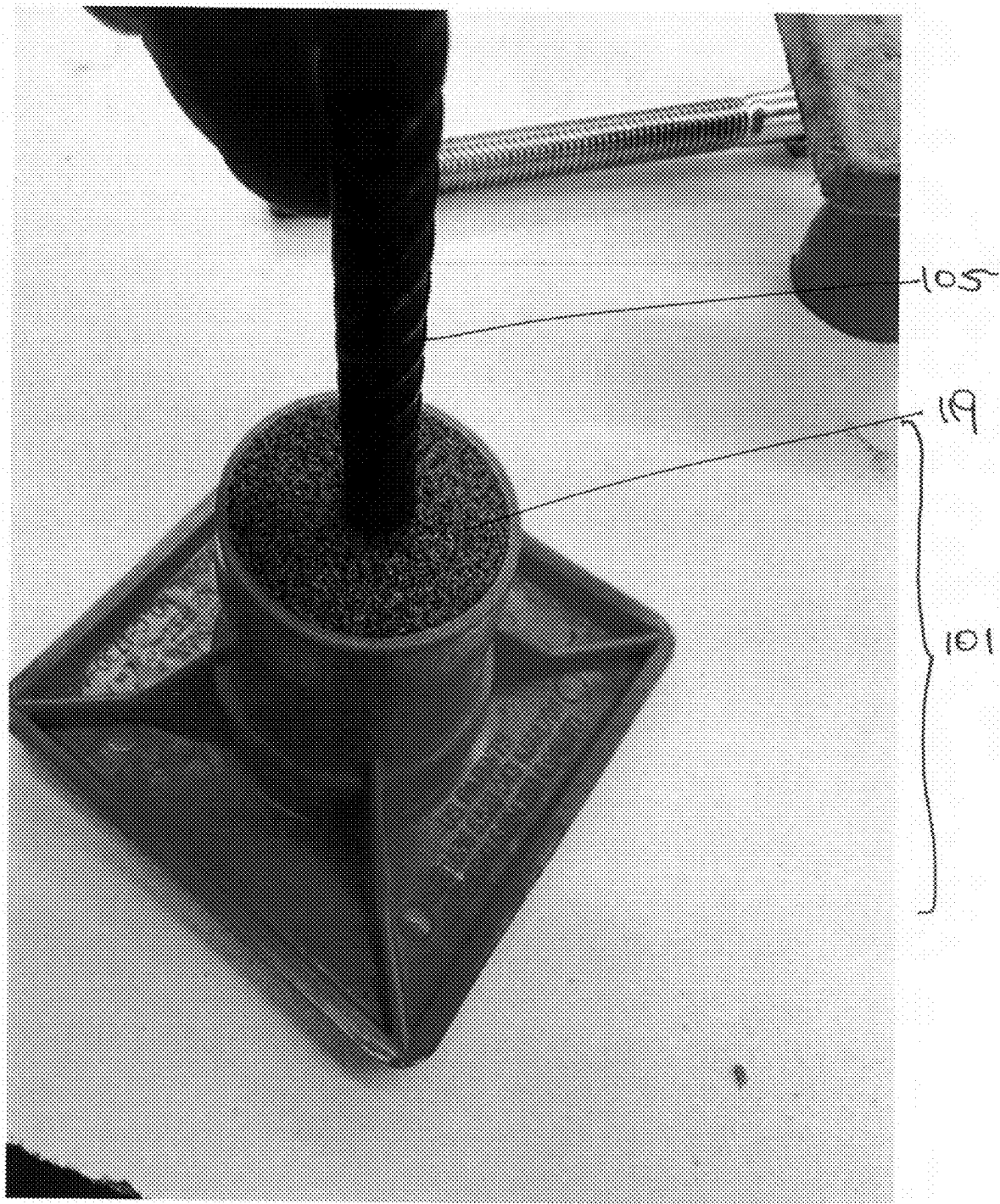


Fig 8



SAFETY CAP RETAINER

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present patent application claims the benefit of U.S. provisional patent application 63/612,213, filed Dec. 19, 2024, the entire disclosure of which is expressly incorporated herein by reference as is fully set forth in this document. Should any word(s) or other content of U.S. Provisional Patent Application No. 63/612,213 be omitted from this non-provisional application, such word(s) or content shall be deemed part of the prosecution history of this non-provisional application and should be taken into account for claim construction purposes unless expressly disclaimed by Applicant in writing.

FIELD AND BACKGROUND OF THE INVENTION

[0002] The present invention relates to a safety device structured to securely hold in place a safety cap at one or both ends of a concrete and masonry reinforcing bar, such as steel reinforcing bars having high tensile strength. Such reinforcing bars are commonly used in the construction industry. The present invention also relates to methods of making and using such safety devices.

[0003] Concrete and masonry reinforcing bars such as, for example, lengths of steel reinforcing rods (known as “rebar”) are commonly used in the construction industry to add tensile strength to other building materials, such as concrete, masonry such as brick and stone, and other building materials which may possess high strength under compression, but low tensile strength. Such tensile reinforcement is today an important feature of buildings, bridges, and other structures, and is often a requirement under local building codes, particularly in seismically active localities. Although rebar is most often made of steel, it may also be made of other materials, such as glass fiber, basalt fiber, carbon fiber or even bamboo. However, in most applications other than specialty construction the cost and thermal expansion properties of these other materials may be less desirable than steel, which has a similar thermal expansion coefficient as concrete. The term “rebar” shall be used herein to mean concrete and masonry reinforcing bars made of any of these materials unless specifically limited to a specific material or set of materials.

[0004] Reinforced concrete usually involves the casting and embedding of lengths of rebar passively in the concrete before the concrete sets to lend the desired tensile strength to the concrete. The ends of the rebar lengths are commonly left uncut and unfinished during the construction process. The rebar ends, which may protrude from the concrete in a vertical, horizontal or angular attitude, constitute a potential real danger for construction workers and others who may trip, or fall upon the rebar end. On construction sites, the presence of such protruding rebar from concrete during the construction of any building, is the source of many accidents, which can range from simple tears in clothing to fatal accidents, including the impalement of workers.

[0005] As a result, protective rebar covers are routinely employed to cover the ends of rebar. A variety of type of rebar caps or covers have been suggested. Many common rebar covers are made of a polyolefin plastic sleeve having an open-ended bore and a hemispherical mushroom-shaped

bulb at the top closed end. These bulbous top caps are effective for preventing cuts caused by contact with the rebar end but are not recommended for use to mitigate against an impalement hazard. To guard against impalement another design is recommended: such rebar covers also comprise a plastic sleeve to receive the rebar end, however these recommended protective rebar covers preferably have a metal (e.g., steel) plate component integrated as part of, or within, the top closed end of the rebar cover. Moreover, the top closed end of the recommended rebar covers comprise an impact surface structured to be oriented at about 90° (e.g., substantially horizontally if the rebar end protrudes vertically) to the orientation of the rebar. The steel plate may be fabricated as part of the polymer body of the protective rebar cover or may be separately inserted within a slot in the top closed end of the protective rebar cover, and is structured to withstand the force of a person falling on the rebar end without the rebar pushing through the top closed end of the cap.

[0006] The polyolefin plastic rebar collars in such protective rebar covers commonly contain a series of 3 or more (usually 4 or more) concentrically arranged flexible plastic fins or ribs facing the interior of the bore through which the rebar end is inserted. These ribs or fins are intended to keep the protective rebar cover longitudinally aligned with the rebar by gripping the sides of the rebar.

[0007] However, in practice one or more of the plastic fins or ribs in the interior void of the bore often break or become bent or inflexible, for example over time or when unintentionally or unknowingly contacted by a worker or a tool, resulting in the protective cover loosening and failing to be securely fastened to the rebar end. In the event a worker were to contact or fall on the end of the rebar, any such insecurely held protective rebar cover can easily become displaced as a result of the force of contact, thereby negating any benefit provided by the rebar cover. This risk is particularly high when the length of the collar of the is relatively short and/or the angle of contact with the rebar cover is oblique.

[0008] Thus, there is a need in the field of construction for an improvement in protective rebar covers to provide the ability to securely retain the rebar covers on protruding rebar with a minimum of time and expense.

BRIEF DESCRIPTION OF THE INVENTION

[0009] The present invention is directed to compositions and methods to securely retain protective rebar covers on the end of protruding rebar during construction projects. Compositions of the invention may include a combination comprising a protective rebar cover and a polymeric foam insert component having a cross-sectional profile approximately matching, but larger than, the bore or interior cross-sectional dimensions of the interior void of the open-ended collar of the rebar cover.

[0010] The polymeric insert component preferably, although not necessarily unless indicated otherwise, comprises a closed cell polyolefin foam of medium hardness and resilience. In this patent application a “medium hardness and resilience” shall include and describe a closed cell polyethylene foam having an average density of about 24-48 kg/m³, a compression recovery of about 95% per minute or greater, a compression deflection of about 38.2 kPa, a tensile strength of greater than about 200 kPa and a water absorption of about 0.03 g/cc or less, and substantially similar

equivalents thereof. An example of a suitable polyolefin foam for use as the polymeric insert component is the closed-cell polyethylene foam sold under the trademark HBR® by Nomaco®, 501 Innovative Way, Zebuyon, NC 27597. This closed cell foam is sold in rods as a cylindrical extrudate having various cross-sectional diameters, or can be custom ordered to have other cross sectional shapes.

[0011] The polymeric insert component preferably has a cross-sectional profile approximately matching and between about 4% or about 6%, or about 10% to about 25%, or about 20%, or about 15% greater in diameter or interior cross-sectional dimensions than that of the interior void of the open-ended collar of the rebar cover. In particularly preferred embodiments the polymeric insert component preferably has a cross-sectional profile between about 4% and about 20%, or between about 5% and about 15%, or between about 6% and about 10%, or about 8% greater in diameter or interior cross-sectional dimensions than that of the interior void of the open-ended collar of the rebar cover.

[0012] In an example of a commonly used protective rebar cover, the internal diameter of the bore or interior void of the open-ended collar is about 1.75 inches. The depth (or axial length) of the bore or void of the open-ended collar is about 1.5 inches from the open end of the collar to the top of the fins or ribs (the portion of the ribs or fins closest to the open end of the collar). The depth of the bore is another 0.5 inches, as measured from the top of the fins or ribs to the closed end of the bore or void.

[0013] The length of the polymeric insert component is preferably equal to or, preferably, less than the depth of the bore. In the example above, the total bore length is 2.0 inches and the polymeric insert component length is preferably about 1.75 inches. This length corresponds to the sum of the depth of the bore of the open-ended collar from the open end of the collar to the top of the fins or rib (1.5 inches) plus half the depth of the bore measured from the top of the fins or ribs to the closed end of the bore or interior void (0.25 inches). Thus, preferably the length of the polymeric insert component is about equal to the total depth of the bore, minus half the depth of the bore measured from the top of the fins or ribs to the closed end of the bore or void.

[0014] In most cases the void or bore within the collar of a protective rebar cover is circular, ovoid, or approximately circular in cross-section. The cross-sectional diameter of a cylindrical polymeric insert component is preferably between about 4% and about 20% greater than the inner diameter of the bore. In one example the interior diameter of the bore is 1.5625 inches, and the cross-sectional diameter of the polymeric insert component is about 1.6875 inches.

[0015] Furthermore, the polymeric insert component comprises a cross-sectional end surface which, when inserted into the bore of the protective rebar cover, faces the open end of the collar of the protective rebar cover. This cross-sectional end surface shall hereinafter be referred to as the “first end surface” and the corresponding end of the polymeric insert component shall hereinafter be referred to as the “first end”. Similarly, the opposing cross-sectional surface of the polymeric insert component shall be referred to as the “second end surface”, and the corresponding end of the polymeric insert component shall be referred to as the “second end”.

[0016] The first end surface of the polymeric insert component is pierced with one or more hole, cut, void, perforation or slit (hereinafter, collectively termed the “slit”)

which is preferably centrally located in at least one dimension on said surface. Thus, for example, if the polymeric insert component is a cylindrical rod, the cross-sectional shape is a circle, and the slit preferably passes through the center of the circle. Although the slit is generally referred to herein as a single hole, cut, void or perforation, in some embodiments the slit may comprise a plurality of such perforations, such as a cross comprising two cuts crossing substantially near the center of the cross-sectional shape. It is thus anticipated that the slit may have at least one length; in such case the edges of the may extend away from the center of the cross-sectional shape, but the midpoint of the slit is preferably centered in substantially the middle of the shape.

[0017] The slit has a depth proceeding through the polymeric insert component in a direction substantially parallel to an axis extending from the first end to the second end thereof. However, the depth of the slit is very preferably shorter than the length of the polymeric insert component itself, and preferably the slit does not pierce the second end surface. In preferred embodiments, the slit has a depth at least equal to the length of the portion of the bore of the open-ended collar at which the ribs or fins are encountered.

[0018] To illustrate using the system described above as an example, the depth of the bore of the open-ended collar from the open end of the collar to the top of the fins or ribs was 1.5 inches, while the total length of the polymeric insert component was 1.75 inches. In this example the depth of the slit, as measure from the first end, is preferably 1.5 inches, leaving the remaining 0.25 inches of the polymeric insert component (to the second end surface) unpierced. In this embodiment, the slit width is 1 inch.

[0019] The polymeric insert component may be pierced, and the slit formed, in any of a number of ways that will occur to a person of ordinary skill in view of the present disclosure, including with an awl, perforations in molded polymeric material, and the like. In a currently preferred method, the slit is made using a press machine having one or more knife of the desired dimensions, corresponding to the desired depth of the slit, and wide enough to receive the rebar end. Each polymeric insert component is positioned under the knife of the press, and the knife lowered into the first end surface at least once to create the slit.

[0020] The polymeric insert component may then be inserted into the bore of the open-ended collar of the protective rebar cover by hand or employing an automated system in which the insert is pressed into the bore until the first end surface of the insert is substantially flush with the open end of the protective rebar cover.

[0021] The present invention is readily scaled up or down depending on the gauge(s) of rebar to be used in the project and the size of the corresponding rebar covers and open-ended collars to be used in conjunction therewith. Imperial/ U.S. rebar sizes are graded in increments of 1/8 inch (3.2 mm), beginning with #2, having a nominal diameter of 0.250 inches and a nominal cross-sectional area of 0.05 square inches, and ending with #18, having a nominal diameter of 2.257 inches and a nominal cross-sectional area of 4.00 square inches. Protective rebar covers can be made to accommodate any of these rebar sizes (and any others used in other countries) and, in view of the present disclosure, polymeric insert components fabricated or customized to fit the bore diameter and depths of the open-ended collars of such rebar covers.

[0022] In use the protective rebar cover containing the polymeric insert component of the present invention is simply positioned over and pushed through the slit onto the rebar end until it stops. The polymeric insert component inhibits side-to-side movement of the rebar within the protective rebar cover, and provides a reinforcing cushion against the fins or ribs within the rebar cover comprising the insert component, lessening their propensity to become bent or broken in use. As a result, the protective rebar cover comprising the insert component is less likely to become loose, displaced or dislodged on the end of the rebar, particularly when inadvertently contacted by a worker. Hence, the polymeric insert component of the present invention provides a clear advantage in reducing worker injury from contacting or falling on the unfinished ends of rebar during construction projects.

[0023] The following additional description is provided for the purpose of further describing aspects, examples, and/or embodiments of the present invention, and is not intended to limit the invention thereto; the invention shall be defined solely by the claims that conclude this application.

[0024] In the present application unless otherwise indicated, each and every range of values (dimensions, physical or chemical properties, time, temperature and the like) stated in this specification, including the claims, are intended to specifically include the entire range defined by the range given to three significant figures, and not just the endpoint (s). For example, a range stated to be 0 to 10 is intended to disclose all whole numbers between 0 and 10 such as, for example 1, 2, 3, 4, etc., all fractional numbers between 0 and 10 to three significant figures, for example 1.50, 2.30, 4.57, etc., as well as the endpoints 0 and 10.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a side view of a protective rebar cover comprising a collar portion and a top closed end comprising an impact surface placed on the end of a vertically protruding segment of rebar.

[0026] FIG. 2 is a top view of a protective rebar cover, turned upside down to show the open end of the collar and the bore formed therewithin. Also visible are four polymer fins or ribs projecting towards the center of the bore.

[0027] FIG. 3 is a top view of a protective rebar cover. Similarly to FIG. 2, the rebar cover is turned upside down to show the open end of the collar and the bore formed therewithin. In this protective rebar cover several broken or bent polymer fins or ribs are shown at the bottom of the bore.

[0028] FIG. 4 shows a side view of a protective rebar cover, turned upside down; a polymeric insert component comprising a slice on the first end surface thereof is shown being placed over the open bore.

[0029] FIG. 5 shows the protective rebar cover of FIG. 4 in which the polymeric insert component is being secured within the open bore of the rebar cover.

[0030] FIG. 6 shows the protective rebar cover of FIG. 4 in which the polymeric insert component has been properly placed within the open bore of the rebar cover with the slice shown on the first end surface thereof.

[0031] FIG. 7 is a top view of the protective rebar cover of FIG. 6 in which the polymeric insert component has been properly placed within the open bore of the rebar cover with the slice shown on the first end surface thereof.

[0032] FIG. 8 is a top view of the protective rebar cover of FIG. 6 in which the end of a segment of rebar is inserted into the slice of the polymeric insert component and pushed down until secure.

DETAILED DESCRIPTION OF THE INVENTION

[0033] The present invention is directed to compositions, combinations, and methods for securing protective rebar covers to the end of an unfinished segment of rebar.

[0034] Protective rebar covers are known in the art. Thus, Australian patent publication AU 2003200539 B is drawn to protective rebar covers containing a compressible block or plug at the inner end of the collar to provide means to absorb the impact force F applied to a construction worker's body.

[0035] French patent publication FR 2459763 A1 discloses a capsule made from soft plastics material and arranged to slide over the ends of steel reinforcing bars used in concrete construction. The capsule has a rounded head and a neck with a curved throat bore to take the steel bar. The capsule is held in position by the neck throat which grips the bar. The end of the capsule has a soft pad made from foam rubber to protect the end of the bore.

[0036] Australian patent publication AU 703323 B describes a protective rebar end cap having a cushioning means located between its upper portion and the free end of the reinforcement bar. The cushioning means may comprise a resilient member such as a dense rubber or a styrofoam block of suitable shape and configuration to fit within the end cap arrangement. The resilient member may be fitted between the upper surface of the end cap and the free end of the reinforcement bar after fitment of the end cap.

[0037] U.S. Pat. No. 4,833,850 discloses an end protector for a reinforcing bar has a one-piece support and a cushion. The support has a cylindrically-shaped receptacle with two arms extending outwardly therefrom and two longitudinal slots that are generally opposed to one another along either side of the receptacle. The arms hold the cushion in position on a side of said receptacle opposite to the reinforcing bar. When an external force is exerted on the cushion, at least part of the force is transferred through said arms to said receptacle, thereby causing the slots to narrow and the receptacle to further tighten onto said rebar. When the external force is removed, the support returns to its initial position.

[0038] U.S. Pat. No. 7,472,522 discloses a rebar protective cover for use on the projecting free end of a concrete reinforcing bar to prevent impact injuries comprising a hollow cylindrical collar, having an open end and a closed end, an overhanging impact head of substantial extent projecting laterally outwardly beyond the closed end of the collar, a bowl-shaped shaping member having the concave surface facing the open end of the collar, and a solid cementitious member occupying the space between the closed end of the collar and the underside of said shaping member. The cementitious member has a surface abutting the underside of the shaping member complementary to the shaping member and adapted to resist impact penetration.

[0039] U.S. Pat. No. 8,776,464 discloses a protective cover including a collar having a first end, a second end, and an impact absorbing portion disposed therebetween. The impact absorbing portion resides solely within the collar. The protective cover further includes a cap disposed at the second end of the collar. The cap has a surface area of about

sixteen square inches. The impact absorbing portion may comprise a seat and a strengthening material. The strengthening material may be disposed above the seat. The strengthening material comprises a high strength polymer configured to increase the impact resistant properties of the protective cap. The strengthening material may, for example, comprise a carbon reinforced polymer, a high strength composite, and/or a high strength ceramic.

[0040] U.S. patent publication No. US 20170175391 discloses a safety apparatus for fitting to a free end of an elongate member, for example a reinforcing bar, has a sleeve portion defining a receiving space for receiving a free end of the member and providing an opening of the receiving space to allow insertion of the member from a first end of the apparatus, and a covering portion which provides a barrier to prevent passage of the member through a second end of the safety device. The safety apparatus may have a gripping part which is resiliently deformable, so that it is resiliently deformed by insertion of the member so as to securely grip metal member, and so that it returns to substantially its non-deformed shape upon removal of the metal member. The receiving space may be tapered.

[0041] German patent publication DE 9004722U discloses a protective cap for reinforced steel rods. To protect a freely protruding end of a reinforcing steel, the protective cap with the open end is pushed onto this reinforcing steel until the end of this reinforcing steel rests approximately on the first of a series of plates in the insertion direction.

[0042] The protective cap is held on the reinforcing steel by clamping areas or by lips located there. In the event of impacts directed in the direction of the longitudinal axis or in the event of a corresponding impact on the protective cap or its head, which is rounded on the entire outer surface, the reinforcing steel pierces one or more of the plurality of plates while simultaneously destroying the impact energy. The head of the protective cap then forms the last barrier.

[0043] However, none of these prior disclosures addresses or solves the problem of ensuring that protective rebar covers are retained on the ends of protruding rebar using a simple and cost-effective retrofitting procedure employing a polymeric insert component. Essentially, the use of such polymeric insert components may be tailored to fit many if not most existing protective rebar covers without the need to redesign an entirely new rebar cover. At the same time, the present disclosure and invention would clearly suggest to the person of skill in the art any of a number of redesigned protective rebar covers which may be fabricated to contain the polymeric insert components of the present invention to ensure that the protective rebar covers are retained securely once they are placed on the end of a segment of rebar.

[0044] As shown in FIG. 1, a combination comprising a protective rebar cover 101 installed on an end of a segment of steel reinforcing bar 105 is displayed in a side view. The exterior of the protective rebar cover comprises a polymeric casing having a collar 103 and a top closed end comprising an impact surface 107. The impact surface 107 at the top closed end comprises a plate-like cap oriented at about 90° (e.g., substantially horizontally if the rebar end protrudes vertically) to the orientation of the rebar 105. The plate-like cap comprises a polymeric casing into which a steel plate (not shown) is inserted or imbedded.

[0045] FIG. 2 shows the underside of a protective rebar cover 101 of the same design as displayed in FIG. 1, including the void or bore 113 formed within the collar 103

and the underside of the impact surface 107 of the top closed end. The collar 103 is substantially cylindrical in shape and comprises a polymeric cylindrical surface 109 defining the hollow bore 113 therewithin.

[0046] As shown in FIG. 2, near the closed end of the bore 113 four fins or ribs 111 are visible projecting generally toward the center of the bore 113. In this rebar cover the fins or ribs 111 are molded as part of the polymeric casing and comprise thin strips of polymer structured to engage the end of the rebar segment 105. In some other rebar covers, the fins or ribs may be manufactured separately. In the embodiment shown in FIG. 2 the ends of the projecting fins or ribs do not touch, and roughly define an open space at the center of the bore near the closed end through which the rebar end may be introduced.

[0047] FIG. 3 shows the underside of a used protective rebar cover 115 of the same design as the protective rebar cover 101 displayed in FIGS. 1 and 2, after field use on the end of a segment of rebar. As can be seen, in the used protective rebar cover of FIG. 3, the fins/ribs 111 are crushed, bent, broken and/or missing, which prevents the protective rebar cover from gripping the rebar end as intended, resulting in the rebar cover only very loosely covering the rebar end when in use, and being susceptible to being displaced completely from the rebar end by environmental factors such as wind or rain, or by accidental contact with the rebar or rebar cover by a worker or item of equipment.

[0048] FIG. 4 shows a side view of a protective rebar cover 101, turned upside down; a polymeric insert component 119 comprising a slit 117 visible on the first end surface thereof 121 is shown being placed by hand over the open bore 113. In this case the polymeric insert component is cylindrical in shape and has a circular first end surface 121. Those of ordinary skill are aware that protective rebar covers may be fabricated to have bores 113 of a shape other than a cylinder, such as, without limitation, a generally cuboid, pyramidal or conical shape, and polymeric insert components may be fabricated to fit these alternative shapes.

[0049] FIG. 5 shows a side view of the protective rebar cover 101 of FIG. 4. In this figure the polymeric insert component 119 is shown in the process of being pushed inside the open bore 113 of the polymeric cylindrical surface 109. Slit 117 is shown in the center of the first end surface of the polymeric insert component.

[0050] FIG. 6 shows a side top view of the combination of the protective rebar cover 101 of FIGS. 4 and 5 with the first end surface of the polymeric insert component 119 placed in position substantially flush with the open end of the polymeric cylindrical surface 109.

[0051] FIG. 7 is a close up of the combination protective rebar cover/polymeric insert component of FIG. 6 showing a close-up of the assembly with the slit 117 shown in the first end surface of the polymeric insert component 119.

[0052] FIG. 8 shows the combination protective rebar cover/polymeric insert component of FIGS. 6 and 7 in which the end of a rebar segment 105 is pushed in place through the slit 117 in the polymeric insert component 119 towards the second end thereof. The resulting assembly retains the protective rebar cover firmly, but removably, on the rebar end with the top closed end of the 107 impact surface retained at the proper, approximately 90° orientation to the axis of the rebar.

What is claimed is:

1. A combination comprising
 - a) a protective cover for a metal reinforcing rod having a substantially hollow elongated collar defining a bore oriented substantially parallel to the axis of elongation of said elongated collar and comprising an open end at a first end of said collar and an integral closed cap comprising an impact surface at a closed second end of said collar, and
 - b) a polymeric insert component securely retained within the bore of the elongated collar, the polymeric insert component having a first end surface substantially flush with the open end of the bore and a second end facing the integral closed cap;
- said polymeric insert component comprising an axial slit or cross formed in said first end surface, sized and structured to receive and securely retain an exposed end of the metal reinforcing rod within the polymeric insert component; said axial slit or cross having a depth less than the length of the polymeric insert component and the depth of said slit or cross extending partly through said polymeric insert component and being oriented substantially parallel to the axis of elongation of said elongated collar.
2. The combination of claim 1 in which the impact surface of said integral cap is substantially perpendicularly to axis of elongation of said elongated collar.
3. The combination of claim 1 in which the polymeric foam insert component has a cross-sectional profile approximately matching, but larger than, the cross-sectional profile of the interior cross-sectional dimensions of the bore of the open-ended collar of the rebar cover.
4. The combination of claim 1 in which the polymeric insert component comprises a closed cell polyethylene foam having an average density of about 24-48 kg/m³, a compression recovery of about 95% per minute or greater, a compression deflection of about 38.2 kPa, a tensile strength of greater than about 200 kPa and a water absorption of about 0.03 g/cc or less.
5. The combination of claim 1 in which the polymeric foam insert component has a substantially cylindrical shape.
6. The combination of claim 1 in which the polymeric foam insert component has a cross-sectional profile substantially similar to, but about 4% to about 25% larger, or about 5% to about 20%, or about 6% to about 15%, or about 6% to about 10%, greater than the interior cross-sectional dimensions of the bore.
7. The combination of claim 1 in which the protective rebar cover contains a series of 3 or more concentrically arranged flexible thin fins or ribs projecting from the sides of the collar, facing the interior void of the bore and structured to grip the sides of the rebar end, said fins or ribs being located near the interior of the closed end of the collar.
8. The combination of claim 7 in which the polymeric foam insert component is about equal to the total depth of the bore, minus half the depth of the bore measured from the top of the fins or ribs to the closed end of the collar.

9. A composition comprising

- a polymeric insert component sized to be securely retained within a bore of a protective rebar cover for an end of a metal reinforcing rod, said protective rebar cover having an elongated collar defining an internal bore oriented substantially parallel to the axis of elongation of said elongated collar and comprising an open end at a first end of said collar and an integral closed cap comprising an impact surface at a closed second end of said collar,
- said polymeric insert component having a cross sectional profile substantially similar to, but about 4% to about 25% larger, or about 5% to about 20% larger, or about 6% to about 15% larger, or about 6% to about 10% larger, or about 8% larger than a cross-sectional profile of said internal bore, and comprising a slit or cross on a first end surface of said polymeric insert component having a depth within said polymeric insert component less than the length of the polymeric insert component.
10. The composition of claim 9, in which the polymeric insert component comprises a closed cell polyethylene foam having an average density of about 24-48 kg/m³, a compression recovery of about 95% per minute or greater, a compression deflection of about 38.2 kPa, a tensile strength of greater than about 200 kPa and a water absorption of about 0.03 g/cc or less.
 11. The composition of claim 9 in which the polymeric insert component is cylindrical in shape.
 12. The composition of claim 9 in which the polymeric insert component is about equal to the total depth of the bore, minus half the depth of the bore measured from the top of the fins or ribs to the closed end of the collar.
 13. The composition of claim 9 in which the polymeric insert component has a length of about 1.75 inches and a diameter about 8% larger than the interior diameter of the bore of said protective rebar cover.
 14. The composition of claim 9 in which the polymeric insert component has a slit or cross length at least equal to the length of the portion of the bore of the open-ended collar measured from the open end of the bore to the position within the bore at which the ribs or fins are encountered.
 15. A method of securing a protective rebar cover on an exposed end of a length of rebar comprising placing the rebar end through the slit or cross in the polymeric insert component of the combination of claim 8 and applying sufficient force to said protective rebar cover to secure the rebar cover to the rebar end.
 16. A method of using the composition of claim 9 comprising securing said composition within the bore of said protective rebar cover so that a first end surface of said polymeric insert component is flush with the first end of the collar, and then placing the rebar end through the slit or cross in the polymeric insert component and applying sufficient force to said protective rebar cover to secure the rebar cover to the rebar end.

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